Material Properties: Aluminium Alloy 6061-T6

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Mechanical properties vs temperature

Temperature (deg C)	Yield strength (MPa)	Tensile strength (MPa)	Elastic modulus (GPa)
20	276	310	68.9
100	262	290	66.0
150	234	250	63.0
200	159	165	59.0

Thermal properties

Property	Value	Unit
Coefficient of thermal expansion (20-100 deg C)	23.6e-6	1/K
Thermal conductivity	167	W/m.K
Density	2700	kg/m3

Note: the fatigue strength at 500 million cycles is 96 MPa at 20 deg C and decreases at elevated temperature.

Design note

At 120 deg C the yield strength is interpolated between 100 and 150 deg C values and should be confirmed by test for safety-critical use.